

Why Drucker–Prager Strip Footing Is Much Harder Than Neo-Hookean in the Static Two-Phase UL-MPM Solver

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1. Question

In the current code base, the strip footing problem with a Drucker–Prager (DP) solid is substantially harder than the same static two-phase updated-Lagrangian MPM solve with a Neo-Hookean solid. In practice this shows up as:

- more Newton iterations,
- more rejected load steps,
- worse behavior of iterative linear solvers,
- stronger sensitivity to load increment size, and
- a need for regularization and line search.

This note explains why that happens from the mathematics of the global problem, rather than from implementation detail alone.

2. The Global Nonlinear System Already Starts Hard

The static two-phase formulation in `src/igampm/MPM/static_twophase_ulmpm.py` solves for nodal displacement and pressure simultaneously. Abstractly, the unknown is

$$\mathbf{x} = \begin{bmatrix} \mathbf{u} \\ \mathbf{p} \end{bmatrix}, \quad \mathbf{R}(\mathbf{x}) = \begin{bmatrix} \mathbf{R}_u(\mathbf{u}, \mathbf{p}) \\ \mathbf{R}_p(\mathbf{u}, \mathbf{p}) \end{bmatrix} = \mathbf{0}.$$

Newton linearization gives

$$\mathbf{K} \Delta \mathbf{x} = -\mathbf{R}, \quad \mathbf{K} = \begin{bmatrix} \mathbf{K}_{uu} & \mathbf{K}_{up} \\ \mathbf{K}_{pu} & \mathbf{K}_{pp} \end{bmatrix}.$$

Even before discussing constitutive law, this matrix is harder than a pure displacement solid mechanics tangent:

1. It is a coupled *mixed* system.
2. The pressure block is not a strong elasticity block; it is mostly generated by mass balance, Darcy mobility, and stabilization.
3. The off-diagonal blocks are essential, not perturbative.

That means the matrix naturally has *saddle-point character*. A saddle-point system is usually much more difficult for Krylov solvers than a well-conditioned symmetric positive-definite elasticity matrix.

3. Why Neo-Hookean Is Comparatively Benign

3.1 Smooth hyperelastic origin

Neo-Hookean response comes from a stored energy density

$$W(\mathbf{F}) = \frac{\mu}{2}(\text{tr } \mathbf{b} - d) + \frac{\lambda}{2}(\ln J)^2 - \mu \ln J,$$

or an equivalent variant. The stress and consistent tangent are derivatives of one smooth scalar potential. In broad terms, that gives:

- a smooth stress update,
- a smooth constitutive tangent,
- no local plastic return map,
- no active-set switching between elastic and plastic branches,
- no yield apex or corner.

As long as the deformation stays in a regime where the material remains strongly elliptic, the displacement block $\mathbf{K}_{uu}$ stays reasonably coercive. It is not perfectly conditioned, but it usually remains a recognizable elasticity operator with predictable spectrum.

3.2 No extra local nonlinear solve

For Neo-Hookean elasticity, one Newton iteration of the global problem requires only:

1. evaluate kinematics at the current particle state,
2. evaluate stress,
3. evaluate tangent.

There is no nested constitutive Newton solve per particle. The global Newton method is therefore handling one nonlinear mechanism at a time: geometric and coupled-flow nonlinearity.

4. What DP Adds Mathematically

DP perfect plasticity changes the structure of the local constitutive problem. At each particle and each global Newton iterate, one no longer has a direct constitutive map

$$\mathbf{F} \mapsto \boldsymbol{\sigma}'.$$

Instead one has a local constrained problem:

$$\boldsymbol{\sigma}' = \mathcal{G}(\text{kinematics}; \text{internal variables}),$$

where $\mathcal{G}$ is defined implicitly through a return map and the yield condition.

4.1 Elastic predictor and return mapping

The local update starts with a trial stress

$$\boldsymbol{\sigma}^{\text{tr}} = \mathbb{C}_e : \boldsymbol{\varepsilon}^{\text{tr}},$$

then enforces the plastic admissibility condition

$$f(\boldsymbol{\sigma}) \leq 0,$$

with equality in the plastic state. In the implemented DP model, the yield function is of the form

$$f(\boldsymbol{\sigma}) = \sqrt{\mathbf{s} : \mathbf{s} + k_f^2} - A_f + B_f p,$$

where

$$p = \frac{1}{3} \text{tr}(\boldsymbol{\sigma}), \quad \mathbf{s} = \boldsymbol{\sigma} - p \mathbf{I}.$$

So the local constitutive law is no longer a simple evaluation. It is a projection onto the yield surface, with a branch change between elastic and plastic response.

4.2 Tangent stiffness softening

The elastoplastic consistent tangent may be written schematically as

$$\mathbb{C}_{ep} = \mathbb{C}_e - \frac{\mathbb{C}_e : \mathbf{a} \otimes \mathbf{b} : \mathbb{C}_e}{\mathbf{b} : \mathbb{C}_e : \mathbf{a} + H},$$

where $\mathbf{a} = \partial g / \partial \boldsymbol{\sigma}$, $\mathbf{b} = \partial f / \partial \boldsymbol{\sigma}$, and H is the hardening modulus. For perfect plasticity, $H = 0$.

This is the first major conditioning hit:

- the rank-one correction subtracts stiffness from the elastic tensor,
- the denominator can become small,
- without hardening there is no positive regularizing term from H ,
- near limit load the effective tangent develops very small eigenvalues.

In physical terms, once a plastic mechanism forms, the solid can continue deforming under only a small load increase. In linear algebra terms, $\mathbf{K}_{uu}$ moves toward singularity.

5. Why DP Is Worse Specifically for Strip Footing

The strip footing problem is not a mild material test. It is close to a collapse mechanism problem.

5.1 Limit-load behavior

Under footing loading, the soil tends toward a bearing-capacity mechanism. As the plastic zone spreads, the load-displacement curve flattens. Symbolically, if the external load factor is λ and the settlement measure is w ,

$$\frac{d\lambda}{dw} \rightarrow 0$$

near collapse.

This has a direct Newton consequence. The Jacobian represents the derivative of the residual with respect to the state. If the equilibrium path is almost horizontal in load space, the Jacobian becomes poorly invertible in the direction of the emerging collapse mode. That is exactly the scenario in which:

- Newton steps become large or erratic,
- tiny perturbations produce large displacements,
- direct solvers still work but require regularization,
- iterative solvers struggle badly.

5.2 Plastic zone propagation changes active sets

The footing problem activates plasticity progressively over many particles. Between two nearby load steps, a set of particles may switch from elastic to plastic. Therefore the global tangent is not only softening; it is also changing *discretely* in structure.

That creates a rougher Newton landscape than Neo-Hookean elasticity, where the tangent changes smoothly with the state.

6. Why the Matrix Is Poorly Conditioned

Several mechanisms stack on top of each other.

6.1 Mixed u - p saddle-point structure

If one eliminates pressure formally, the Schur complement is

$$\mathbf{S} = \mathbf{K}_{pp} - \mathbf{K}_{pu}\mathbf{K}_{uu}^{-1}\mathbf{K}_{up}.$$

Good conditioning requires both $\mathbf{K}_{uu}$ and $\mathbf{S}$ to be healthy. In the static two-phase problem:

- $\mathbf{K}_{uu}$ deteriorates when plasticity spreads,
- $\mathbf{K}_{pp}$ may be weak if mobility is small,
- the coupling blocks remain strong,
- therefore $\mathbf{S}$ becomes hard to approximate well.

This is why generic preconditioners such as Jacobi or ILU often disappoint. They do not capture the actual mixed physics of the operator.

6.2 Loss of coercivity in the displacement block

For Neo-Hookean elasticity, $\mathbf{K}_{uu}$ is roughly an elasticity-plus-geometric stiffness matrix. For DP perfect plasticity, it becomes an elastoplastic tangent that is much softer in the mechanism direction. The smaller the smallest singular value of $\mathbf{K}_{uu}$, the larger the condition number

$$\kappa(\mathbf{K}_{uu}) = \frac{\sigma_{\max}(\mathbf{K}_{uu})}{\sigma_{\min}(\mathbf{K}_{uu})}.$$

Near collapse, $\sigma_{\min}(\mathbf{K}_{uu})$ can become very small.

6.3 Non-associativity can destroy symmetry

If the flow rule is non-associative, then

$$\frac{\partial g}{\partial \boldsymbol{\sigma}} \neq \frac{\partial f}{\partial \boldsymbol{\sigma}}.$$

Then the consistent tangent is generally non-symmetric:

$$\mathbb{C}_{ep} \neq \mathbb{C}_{ep}^T.$$

Once assembled into the global matrix, this matters immediately:

- methods relying on symmetry become less appropriate,
- preconditioner design becomes harder,
- spectral intuition from SPD matrices no longer applies.

Even when the global matrix is solvable, it is now harder to solve *robustly*.

6.4 Perfect plasticity gives no hardening buffer

Hardening acts like a stabilizer in the denominator of the return-map consistent tangent. In the schematic formula above, the denominator contains H . With

$$H = 0,$$

the model has no constitutive buffer against mechanism formation. That does not make the model wrong, but it makes the linearized problem much less forgiving.

6.5 Apex and pressure-sensitive coupling

DP is pressure-sensitive:

$$f = f(p, q).$$

That means volumetric and deviatoric modes are intrinsically coupled. Under footing loading, mean stress changes are important, so the plastic response is not a simple deviatoric correction. In contrast, Neo-Hookean elasticity does not have a yield surface whose normal rotates sharply as the stress state evolves.

Also, cone-like yield surfaces have an apex-related sensitivity. Even with the smoothed implementation used here, the local constitutive map is less regular than a hyperelastic law.

7. Why Newton Needs More Protection

When the tangent is soft, mixed, and branch-dependent, raw Newton becomes brittle. Suppose the exact Jacobian is $\mathbf{K}$ and the residual is $\mathbf{R}$. Then the Newton step is

$$\Delta \mathbf{x} = -\mathbf{K}^{-1} \mathbf{R}.$$

If $\mathbf{K}$ is ill-conditioned, a moderate residual can generate a very large step. That is why the code benefits from:

- load stepping,
- step rejection,
- line search,

- direct regularization,
- dual convergence checks on both residual and step size.

The key point is that a small residual alone is not enough. For an ill-conditioned matrix, one may have

$$\|\mathbf{R}\| \text{ small}, \quad \|\Delta\mathbf{x}\| = \|\mathbf{K}^{-1}\mathbf{R}\| \text{ not small.}$$

That is exactly why the added check on $\|\Delta\mathbf{x}\|_\infty$ matters for the DP strip footing case.

8. Why Direct Solvers Work Better Here

The numerical experiments in this repository show that the current DP strip footing path is robust with:

- direct solve,
- small diagonal regularization,
- careful load stepping.

This is mathematically consistent with the structure above.

8.1 What the regularization does

Adding

$$\varepsilon\mathbf{I}$$

to the linear system shifts the spectrum away from zero. If $\mathbf{K}$ has small or nearly zero eigenvalues, then

$$\mathbf{K}_\varepsilon = \mathbf{K} + \varepsilon\mathbf{I}$$

has improved invertibility. This does not cure the underlying constitutive softness, but it prevents the solve from collapsing on a nearly singular direction.

8.2 Why Krylov methods struggle

GMRES, MINRES, and related iterative methods are strongly affected by:

- clustered near-zero eigenvalues,
- non-normality,
- poor Schur complement approximations,
- non-symmetry from non-associative plasticity,
- changing active sets between Newton iterations.

So the issue is not that iterative methods are universally bad. The issue is that this particular operator is a difficult monolithic mixed elastoplastic operator, and simple preconditioners do not reproduce its block physics accurately enough.

9. Why Neo-Hookean Does Not Show the Same Severity

The comparison is clearer if one lists the main differences.

Feature	Neo-Hookean	Drucker–Prager
Constitutive type	hyperelastic	elastoplastic return map
Local branch switch	no	yes
Yield surface	none	yes
Perfect plastic softening effect	no	yes
Consistent tangent regularity	smooth	less smooth
Potential symmetry	often better behaved	can be non-symmetric
Near-collapse mechanism	weaker in this test	central feature
Conditioning of $\mathbf{K}_{uu}$	moderate	much worse
Sensitivity to load increments	lower	much higher

So the correct conclusion is not merely “DP is nonlinear.” Neo-Hookean is also nonlinear. The real difference is:

DP introduces pressure-sensitive plastic admissibility, tangent softening, active-set changes, possible non-symmetry, and near-mechanism behavior inside an already difficult mixed u - p system.

10. Consequences for This Code Base

For the present implementation, the mathematical analysis points to the following practical conclusions.

10.1 Current robust path is reasonable

The present successful path

`direct + small regularization + load stepping`

is not a hack in the pejorative sense. It is a numerically reasonable response to a nearly singular mixed elastoplastic problem.

10.2 Dual convergence checks are necessary

Because the Jacobian may be ill-conditioned, residual reduction does not imply that the Newton step is small. Requiring both

$$\|\mathbf{R}\|_{\infty} \leq \text{tol}_R, \quad \|\Delta \mathbf{x}\|_{\infty} \leq \text{tol}_x$$

is mathematically justified.

10.3 Better iterative performance requires physics-aware preconditioners

If one wants iterative solvers to compete with direct solve, the preconditioner must approximate the block structure more faithfully, especially the Schur complement:

$$\mathbf{S} = \mathbf{K}_{pp} - \mathbf{K}_{pu} \mathbf{K}_{uu}^{-1} \mathbf{K}_{up}.$$

In practice that means one or more of:

- stronger block factorization preconditioners,
- better Schur complement approximations,
- constitutive regularization or hardening,
- arc-length or continuation methods near limit load,
- solver strategies specialized for mixed plasticity systems.

11. Bottom Line

The DP strip footing problem is much harder than the Neo-Hookean case for structural mathematical reasons:

1. the global problem is already a mixed u - p saddle-point system,
2. DP perfect plasticity softens the displacement block,
3. near bearing-capacity failure the tangent approaches singularity,
4. active-set changes and non-associativity make the tangent less benign,
5. therefore the global matrix becomes badly conditioned and harder to solve.

Neo-Hookean does not remove all nonlinearity, but it preserves a smooth energy-based constitutive map and avoids the plastic mechanism physics that degrade the tangent so severely. That is why the difference in solver behavior is large in practice and fully expected from the mathematics.